LIBBY — PATIENT/CAREGIVER TRACK

osteosarcoma-mets-dll3-h7r2

Plain-language summary for patient and caregiver

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What this page is

You (or your clinician) asked Libby a narrow question: in metastatic osteosarcoma where there's a hint that DLL3 might be present, what treatments would target DLL3? You also said you'd consider high-risk options if the upside is meaningful, and the board read this report through that lens.

One thing matters before anything else. The DLL3 signal you gave us was at the RNA level. RNA tells us the gene is being read inside the cell. It does not tell us the DLL3 protein is actually sitting on the outside of the cell, which is what the new drugs need to grab onto. That distinction shapes the rest of this page. So the recommendations come in two parts. First, a confirmatory test that everyone agreed should happen. Second, a single treatment option that only opens up if that test comes back positive.

The page is also scoped tightly. It only covers drugs that target DLL3, because DLL3 was the question. Plenty of other treatments exist for relapsed osteosarcoma, and those are a conversation for your oncologist, not this report.

What we know about your cancer

You're in your late teens or twenties with metastatic (stage IV) osteosarcoma, a cancer that started in bone. The board assumed you'd had standard frontline chemotherapy (the MAP regimen: methotrexate, doxorubicin, cisplatin), since that's the default for this disease, but your inputs didn't say how you responded or whether you've had anything since. Your day-to-day function looks like ECOG 1, which roughly means you're up and around but tire more easily than you used to. The single feature driving this report is the DLL3 RNA signal, with the caveat from the section above.

What you told us matters most

You said you accept high-risk, high-reward options when the possible upside is meaningful. You didn't name any toxicities you wanted to veto outright, and you flagged a clear preference for clinical trials over off-trial experimental use. The board took those signals seriously. They're the reason a first-in-disease trial option even made it onto the page.

The first step everyone agreed on

Before any treatment decision, get a DLL3 protein test on the tumor. The specific test is an IHC stain that uses an antibody called SP347. In most cases your oncologist can order it on stored tissue from a previous biopsy. If there isn't enough archived tissue, a fresh sample works too. The test itself isn't toxic. Turnaround is roughly one to three weeks. The cost is trivial compared to a single treatment cycle.

All five reviewers put this in the rank-1 slot, and nobody dissented. The result decides whether the option below is on the table at all. If the test comes back positive, you have a real choice to discuss. If it comes back negative, the DLL3-directed pathway closes and the conversation shifts to standard relapsed-osteosarcoma options that your oncologist can take you through.

The option the board considered

Option 1 — Tarlatamab via clinical trial NCT06788938

This option is conditional on a positive DLL3 IHC. If the IHC is negative or below the trial's threshold, the option is off the table, and this page has no further treatment recommendations within scope.

When the test is positive, the board recommended tarlatamab as the lead option.

So what is it? Tarlatamab is a "bispecific T-cell engager." You get it as an infusion every two weeks. The drug works like a molecular clip: one end grips DLL3 on the cancer cell, the other end pulls in one of your own T cells to attack it. It's currently approved for small-cell lung cancer. NCT06788938 is a "basket" trial, which means it accepts any DLL3-positive tumor (osteosarcoma included), provided the IHC confirms DLL3 is on the cell surface at high enough levels.

What might the upside look like? In small-cell lung cancer, where this drug is already approved, a recent randomized trial showed tarlatamab cut the risk of death roughly in half compared to chemotherapy. About 40 out of 100 patients had measurable tumor shrinkage. Whether any of that carries over to osteosarcoma is exactly the unanswered question this trial is designed to answer. I can't translate the SCLC numbers cleanly into an osteosarcoma expectation here, because no one has published that data yet and it would be misleading to pretend otherwise.

On the risk side, about half of patients get a reaction called "cytokine release syndrome." It's a flu-like immune flare; most cases are mild and roughly 1 in 100 are severe. About 1 in 10 patients have neurologic side effects. The first cycle requires inpatient monitoring, so you'd be in hospital while the step-up doses are given.

The board wasn't unanimous on this one. Four reviewers endorsed it in the positive-IHC scenario (the risk-taker, the advocate, the consensus reviewer, and the conservative). The conservative's earlier worry was specifically about treating an unconfirmed target; once IHC locks the target down, that worry goes away. The critic still dissented even with a positive IHC, on a single specific point: no osteosarcoma patient has been published as receiving tarlatamab yet. Whoever takes this trial is in the very first cohort. That's a real thing to weigh, not a footnote.

If the IHC comes back negative, the option closes entirely. There's no backup within scope of this page, because the page was scoped to DLL3. Standard 2L+ options for relapsed osteosarcoma exist and are well established. Your oncologist is the person to take you through them.

What the board could not figure out

A few things the inputs didn't tell us, which would sharpen the plan:

- Was DLL3 measured at the protein level (IHC) or only at RNA? The
- What treatment have you actually had? The recommendations assume
- What's your day-to-day function (performance status)? Most trials
- Where do you live, and how reachable are academic cancer centers? The

Questions to ask your oncologist

1. Can we order DLL3 IHC on my tumor using the SP347 antibody? That has
2. If the IHC comes back positive, what's the realistic path to NCT06788938
3. If the IHC is negative, what 2L+ options do you usually consider for
4. If we go with tarlatamab, how long does it usually take from IHC
5. What's your hospital's protocol for the side effects, especially
6. Have I already had any of the standard 2L+ osteosarcoma options? What
7. Is there anything in my labs or imaging that would change which trials
8. If the tarlatamab trial turns out to be unreachable for me, whether

Where to read more

- Clinician summary — the technical version of this page.
- Trial table — the trials Libby reviewed.
- Evidence list — the clinical and lab studies cited.
- Tumor-board transcript — what each reviewer actually said.
- Recommendations detail — full structured tables.

Sources

The recommendations cited the following published reports and clinical-trial records:

- Ahn et al. *N Engl J Med* 2023 — tarlatamab DeLLphi-301 in SCLC (PMID 37861218)
- Mountzios et al. *N Engl J Med* 2025 — tarlatamab DeLLphi-304 in SCLC
- Yao et al. *Oncologist* 2022 — DLL3 as a therapeutic target review
- Wang et al. *J Exp Clin Cancer Res* 2018 — Notch3/DLL3 axis in osteosarcoma
- Trial registry NCT06788938 — UCCC-01 / UCLA L-10 tarlatamab basket